FIRO – Wildfire Detection Using Edge Devices and MobileNetV2 Lite

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by

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Project Detail

Type of project ☐ Development ☐ Research ☐ R&D				
Area of specialization: **Computer Vision**				
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Abstract

Wildfires are increasingly becoming a danger to forest ecosystems, biodiversity, and human lives especially in the climate-prone areas. Current methods of wildfire detection system are heavily dependent on satellite imagery and reporting methods, which lack expediency of response and real-time functionality. FIRO introduces a lightweight edge-detection system used to detect wildfire in real-time using low-power devices. The suggested system is based on a MobileNetV2 Lite model that was optimized with the help of the transfer learning technique and the quantization to INT8 based on the TensorToolkit to detect fire scenes and non-fire ones with images only. This optimized model is implemented on a Raspberry Pi and it is combined with cloud based logging system and a web-based monitoring dashboard. Experimental analysis has shown that classification accuracy is 97.5% with low inference latency which is practical in continuous deployment. The findings reveal that lightweight deep learning models paired with edge computing can be used as the cost-effective and scalable solution to the early wildfire detection in resource-constrained settings.

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Chapter 1

Introduction

1.1 Introduction

Environmentalists have warned that Earth's ecological systems are approaching critical thresholds, leading to abrupt and irreversible environmental changes. The increasing temperatures of the world, extended dry seasons, glaciers melting, massive deforestation of land, etcetera have made forest ecosystems across the world more vulnerable [1]. Among such issues, forest fires that go out of control have become one of the most devastating environmental threats that lead to massive destruction of vegetation and biodiversity, soil fertility, and ecosystem stability. Therefore, forest resource management and the prevention of wildfires damage has become one of the most important environmental problems of the 21st century [2]. Figure 1.1 illustrates the December 2025 Neelum Valley forest fire incident [3].

Despite being a global threat, the scale of wildfire emergence is disproportionally large and severe in developing and underdeveloped countries because of the resources available, the insufficient environmental regulation, and technological infrastructure. Pakistan and Azar Jammu and Kashmir are the most affected areas with countries like India, Pakistan, being highly vulnerable. In Pakistan, poor environmental policies, logging, inadequate forest surveillance systems, lack of finance, and high availability of modern monitoring technology have contributed greatly towards forest degradation. Consequently, this

Figure 1.1: December 2025 Neelum Valley forest fire, AJK Pakistan. Image source: Discover Pakistan News Channel (actual incident photograph)

has continued to pose a consistent threat to forest ecosystems in the North of Pakistan and Azar Kashmir due to deforestation and frequent wildfires with low potential to detect and respond in time. Current forest surveillance systems tend to be expensive, less densely distributed and cannot collect data and predict new trends in real-time, and this is why successful maintenance of the wildfires is exceptionally problematic.

With the introduction of machine learning and high-level methods of data acquisition, the detection and monitoring of wildfires have changed in many ways. Contemporary deep-learning based systems together with edge-computing devices and RGB and thermal cameras allow detecting smoke and fire outbreaks earlier and with greater accuracy than traditional satellite-based or manual surveillance devices. Such AI-based mechanisms are capable of analyzing any visual images, patterns of smoke spread, fire patterns in real time including signed conditions like low visibility or at night. It offers scalable and cost-effective methods to early wildfire detection, especially in hazardous forest regions of Pakistan and Azad Kashmir through the deployment of optimized neural networks, including Convolutional Neural Networks, on a low-cost edge hardware.

In the last three years, the cases of forest fires were rising and becoming more brutal in Pakistan, especially in the forested regions in the north. The middle of December is when the peak wildfire season usually starts, and lasts a period of about 25 weeks. The high-confidence Visible Infrared Imaging Radiometer Suite (VIIRS) fire alerts were 2,214 in April 29, 2024, to April 28, 2025, spread across the country [4], which has been noticed to increase the intensity of wildfire occurrence in the past years. It is noteworthy that this amplifies how vulnerable the forest ecosystems in Pakistan have become and how there is an urgent need to have effective early detection systems.

Wildfire monitoring in Pakistan currently depends mainly on public reporting, manual watchtowers, and satellite imagery. Satellites have a wide coverage area but have issues of temporal delays, interference by the clouds as well as a low spatial resolution rendering them ineffective in detecting fires early. The manual observation approaches are cumbersome and cannot be applied where there are mountains. Figure 1.2 illustrate Fire Alerts in Pakistan between Jan-Apr 2025. [5]

In 2025 alone, Pakistan recorded 966 high-confidence VIIRS fire alerts [6], as illustrated in *Figure 1.1.* the year 2024 recorded the highest number of wildfire incidents, with 1,905 alerts. These statistics underscore the escalating risk of forest damage due to wildfires and reinforce the need for real-time, low-cost, and automated wildfire detection systems capable of supporting timely intervention and forest conservation efforts. Some of recent forest fire incidents in Pakistan and Azad Jammu Kashmir are in Table 1.1.

Recent developments on the Artificial Neural Networks (ANNs) and deep

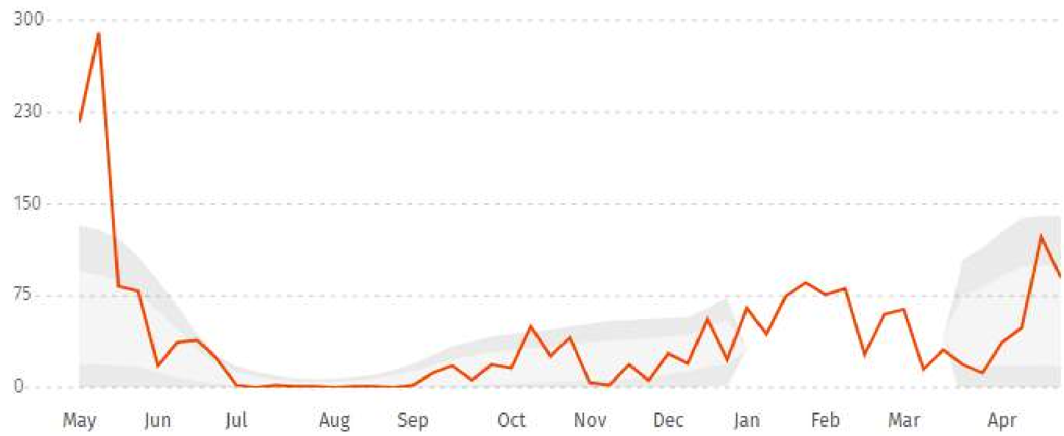

Figure 1.2: Fire alerts in Pakistan Jan-Apr 2025(Image Source Global Forest Watch)

learning have made it possible to use computer vision to detect wild fires automatically. Nonetheless, most of the high performance models rely on computationally intensive architectures and cloud computing, which make them less applicable in the remote and resource limited environment. FIRO is an edge-based wildfire detector that is capable of running on low-power computers but at high detection rates.

Table 1.1: Wildfire incidents in Pakistan and Azad Kashmir (2020-2024)

Year	**Location**	**Area (ha)**	**Key Impacts**	**Cost**	**Source**
2020	Sherani Forest	8,000+	30% olive trees destroyed Long-term soil degradation Economic impact on local farmers	$2.7M	Dawn News [7]
2022	Margalla Hills	500+	Biodiversity damage Military intervention Air quality crisis	$2.1M	Express Tribune [8]
2024	Gilgit-Baltistan	600+	Threat to glacial ecosystems Tourism revenue loss Water source contamination	$1.2M	SUPARCO [9]

1.2 Goals and Objectives

1.2.1 Goals

The primary goal of the FIRO project is to design and implement a low-cost, lightweight, and real-time wildfire detection system capable of operating on resource-constrained edge devices. The project aims to leverage deep learning–based computer vision techniques to enable early detection of forest fires using Cameras, thereby supporting timely intervention and reducing environmental and economic damage in forest regions of Pakistan and Azad Jammu and Kashmir.

1.2.2 Objectives

The specific objectives of this project are:

(i) To design and implement a real-time, edge-based wildfire detection system that autonomously captures images using a camera-mounted Raspberry Pi and performs on-device inference without reliance on cloud-based processing.

(ii) To obtain credible wildfire detection with high classification accuracy and low inference latency to facilitate continuous, twenty-four-hour environmental monitoring under varying lighting conditions.

(iii) To create a web-based monitoring dashboard that utilizes a cloud backend to visualize detection outcomes, log wildfire events, enable public fire incident reporting, and transmit only compact prediction metadata to reduce network bandwidth usage.

1.3 Problem Statement

Forest fire monitoring in Pakistan and Azad Kashmir is largely limited, with only a small portion of forest areas under active surveillance and heavy reliance on manual reporting. This often leads to delayed detection and slow response, especially in remote regions. To address this, the project presents a low-cost, edge-based wildfire detection system for early and reliable fire detection in real-world environments.

1.4 Assumptions

To ensure a realistic and feasible evaluation, the system is designed under the assumption that, during normal operation, the deployed cameras capture images with sufficient clarity to distinguish between fire and non-fire scenes. Adequate lighting, visibility, and stable camera placement are considered necessary for reliable visual feature extraction and accurate classification. It is also assumed that the visual patterns present in the training data—such as flames, smoke behavior, vegetation, and environmental context—are broadly representative of real-world wildfire scenarios, allowing the system to generalize effectively during deployment.

In addition, the system assumes continuous and stable operation of the edge hardware, supported by a reliable power supply to enable uninterrupted real-time monitoring. Basic network connectivity is presumed to be available for synchronizing detection results, enabling remote monitoring, and delivering alerts through the web interface, even though core processing occurs locally. Finally, it is assumed that extreme environmental conditions such as heavy fog, intense rainfall, or complete darkness occur infrequently, as such conditions may degrade image quality and reduce detection reliability. These assumptions collectively define the operational boundaries within which the system is expected to perform effectively.

1.5 Project Scope

The scope of this project is to develop a complete end-to-end wildfire monitoring pipeline, encompassing data preparation, deep learning–based model development, edge deployment, and a web-based monitoring and reporting interface. The system is designed to operate in real time, enabling timely situational awareness for both relevant authorities and the general public, while maintaining reliable performance under normal environmental conditions. The project emphasizes a standalone and deployable architecture in which inference is performed at the edge, with only essential detection metadata transmitted to the cloud for visualization and logging. Advanced extensions such as multi-sensor fusion, satellite-based integration, and large-scale predictive analytics are deliberately excluded to keep the system within a manageable scope and to focus on validating the effectiveness of an efficient, edge-centric wildfire detection solution for early warning applications.

Chapter 2

Requirement Analysis

2.1 Literature Review

The topic of wildfire detection has received ample research attention with the rising number and intensity of forest fires across the world. The major sources of the early detection were satellite imagery, lookout towers, and manual reporting. Despite the advantages of satellite-based solutions in offering extensive coverage, they are usually characterized by extended revisit times, interference by clouds, and take time to respond which hamper the capabilities of satellite-based solutions in detecting and responding to wildfires in their initial stages. In Chan et al., an extensive survey of wildfire monitoring technologies was introduced and offered by the authors, pointing to the shortcomings of classical satellite and manual methods and the opportunities of further development of solutions based on vision and IoTs [10]. The research found the following challenges as essential to practical systems of wildfire monitoring: real time processing, power efficacy, and deployment scalability.

Due to the growing popularity of deep learning, convolutional neural networks (CNNs) have been popularly used to detect wildfires and smoke. InceptionV3 model that was used to achieve 86 per cent true positive detection in a CNN-based smoke detection system proposed that is very good for us by Govil et al. [11] . Nevertheless, the detectivity reported to be as slow as 17.75 minutes and the absence of edge case deployment assessment makes it less suitable in responding to wildfires in real time.

In order to minimize the level of complexity during computation, some studies have investigated the lightweight architectures. FireTech [12] used hierarchical knowledge distillation to reduce the size of DenseNet201 to MobileNetV3Small to detect wildfires on a drone by 93.36 and with fewer false alarms. Although the efficiency was enhanced, the methodology was restricted to classifying the static images only and it did not cover continuous edge-based monitoring.

Multi- modal and satellite based methods are also investigated. Seydi et al. proposed Fire-Net, a two-stream CNN that combines both RGB and thermal satellite images, with an accuracy of 97.35 percent unified with these two streams and inputs using images of the fire [13]. However, the real-time will be limited to Landsat-8 imagery due to a 16 days revisit cycle. Equally, J. R. Martınez-de Dios et al. [14] incorporated rule-based approaches with a two-channel CNN and achieved very high accuracy, but was tested in Portuguese forest conditions only, and thus, it remains unclear whether it could be generalized to other locations.

More recent literature has examined edge deployment advanced architectures and techniques. Mahmoudi et al. [15] among other hybrid CNN Vision Transformer models that were optimized by pruning and quantization has stated that it gave 98 percent accuracy with lower memory consumption. Although advantageous, the study was based on a fairly limited set of data and lacked in-depth cross-dataset verification.

The issue of balancing data has also been discussed in the previous works. Jiao et al. [16] tested CycleGAN-driven data augmentation using DenseNet to obtain a high accuracy of 98.27 percent in their study [16]. Nevertheless, generated samples realism and inference performance on embedded devices was not examined.

Wildfire detection systems that are implemented by drones have attracted attention because of their flexibility. A multi- sensor architecture (RGB, thermal, humidity, temperature) has been suggested by Lelis et al. [17] that showed to be effective with regard to predicting spatial risk. However, reliance on server processing and lack of scalability are still issues to the scalability of long-term operation.

Segmentation models like FireFormer built upon transformers basis multi-channel segmentation work achieved good segmentation performance through the incorporation of CNNs and attention model [18] are energy efficient systems that require close interdependence with the cloud and network stability.

A meta-analysis that was done of 35 works on the topic of wildfire detection to check the fire in forest by Ramos et al. claims that current research is wholly dominated by CNN-based models such as the YOLO or MobileNet variants of this algorithmic type, these models and algorithm was adopted in Rmos to check the detection as reported by Ramos et al. [19]. The authors have noted the unavailability of latency measurements in real world and scanty testing on resource-constrained hardware as major research gaps.

Early research on wildfire detection primarily relied on conventional convolutional neural networks and vision-based systems deployed on fixed camera infrastructures. Initial works demonstrated the feasibility of using deep learning for smoke and fire detection from live camera feeds, highlighting reduced false positives and early detection capabilities in controlled environments [20]. Subsequent studies improved robustness by addressing challenging environmental conditions such as fog and low visibility through energy-efficient CNN architectures suitable for IoT devices [21]. Architectural enhancements, including dilated convolutions and data augmentation using generative adversarial networks, further improved detection accuracy while attempting to balance computational complexity [22, 23]. Ensemble-based approaches combining multiple detectors such as YOLO, EfficientDet, and EfficientNet achieved real-time performance but introduced higher computational overhead, limiting their applicability to resource-constrained edge devices [24].

As research progressed, several studies focused on improving detection accuracy and reducing false alarms through advanced architectures and learning strategies. Encoder–decoder frameworks and segmentation-based models enabled pixel-level fire localization, often by fusing RGB and thermal imagery, though they were constrained by limited labeled datasets [25]. Attention-based dual-channel systems and continual learning methods demonstrated significant reductions in false positives and improved adaptability across datasets, albeit at the cost of increased training complexity [26, 27]. Educational and experimental studies explored spatial–temporal fusion to enhance detection performance in video streams, while extended GAN-based evaluations investigated robustness to illumination and noise variations [28, 29]. Despite these advances, many approaches remained computationally intensive and unsuitable for real-time deployment on low-power platforms.

Recent studies have increasingly emphasized lightweight models, explainability, and edge deployment feasibility. Knowledge distillation techniques have been employed to transfer knowledge from large teacher models to compact student networks, enabling efficient deployment on drones and embedded devices [30, 31]. Explainable AI frameworks integrated visualization techniques such as Grad-CAM, LIME, and SHAP to improve transparency and trustworthiness of wildfire detection systems, though often requiring extensive computational resources [32]. Parallel research explored sensor-based AI systems and UAV-based monitoring to enhance situational awareness, integrating environmental data and transformer-based architectures for global context modeling [33–35]. More recent works investigated robotic platforms, vision transformers, and hybrid edge AI systems with pruning and quantization to reduce energy consumption and inference latency [36–38]. While these approaches demonstrate promising results, the need for a cost-effective, lightweight, and real-time wildfire detection system optimized for low-power edge devices remains a critical research gap, which this project aims to address.

In general, available literature indicates high detection rates with frequent use of computationally expensive models, online processing or latent data. On the other hand, the suggested FIRO system concentrates on one lightweight MobileNet V2 Lite system that is programmed to operate in real-time to identify wildfire. This architecture focuses on low latency, smaller model size, and can be implemented on the available forest conditions of limited resources.

Table 2.1: Comparative analysis of wildfire detection research (2019–2025)

Year	**Study**	**Technique / Model**	**Dataset (Type & Size)**	**Accuracy / Metrics**
2019	S. Parks et al. [20]	InceptionV3 CNN	Live camera frames (HP-WREN & ALERT)	Detect smoke in ∼15 min; <1 FP/day
2019	Y. Zhang et al. [21]	Modified VGG-16	Foggy smoke dataset + SmokeDB	∼96% accuracy; 0.9% FP
2020	B. Ko [22]	Dilated CNN	∼70k frames	∼99% accuracy; P/R ∼97–98%
2020	P. Barmpoutis et al. [23]	DenseNet + CycleGAN	4.9k non-fire, 1.4k fire + synthetic	98.27% acc; F1=98.16%
2021	Y. Zhao et al. [24]	YOLOv5 + EfficientDet + EfficientNet	BowFire, VisiFire, ForestryImages (∼10k)	97.4% acc; 45 FPS
2021	M. Cheng et al. [25]	SegNet + ResNet50	Fire-Net (1,050 labeled images)	97.35% acc; IoU=90.7%
2022	J. R. Mart ́ınez-de Dios et al. [26]	Dual-channel CNN + Attention	CICLOPE watchtower data	99.7% acc; FP=0.12–0.24%
2022	S. Khan et al.S. Khan et al. [27]	VGG16 / Xception + LwF	MODIS, Sentinel, Kaggle (∼6.9k)	96.89% acc
2023	A. Singh [28]	Reduced VGG + temporal fusion	∼2k images	97.3% (fusion)
2023	P. Barmpoutis [29]	DenseNet + GAN	Kaggle + custom	96–97% acc
2024	J. Lee et al. [30]	DenseNet201 → MobileNetV3	Hierarchical wildfire dataset	93.36% acc; AUC=0.96
2024	M. Ahmad et al. [32]	MobileNetV3 + XAI	Kaggle + Mendeley (∼30k)	99.91% acc
2024	S. Raza et al. [33]	FFNN + LoRa WSN	Real forest sensors	Qualitative validation
2024	L. Zhang et al. [34]	CNN + Transformer	FLAME dataset	IoU=73.1%; F1=84.5%
2024	J. Alonso et al. [35]	YOLO + ML risk models	UAV missions	<10 s latency
2025	R. Silva et al. [36]	Classical CV	Onboard drone video	Qualitative
2025	H. Chen et al. [37]	ViT-base	∼10GB images	96.1% acc
2025	Y. Kim et al. [31]	MobileNetV3 KD	Hierarchical dataset	93.36% acc
2025	A. Ahmed et al. [38]	Pruned CNN + ViT	∼1.5k images	98% acc; 6× less energy

2.2 Stakeholders List

Table 2.2 presents the key stakeholders involved in the FIRO wildfire detection system and outlines their respective roles and responsibilities. It identifies the primary users, administrators, and system components that interact with the system during operation and maintenance. This table helps clarify stakeholder involvement to ensure effective system usage, monitoring, and decision-making.

Table 2.2: Stakeholders of the FIRO Wildfire Detection System

Sr.#	Stakeholder	Role	Interest / Responsibility
1	Forest Department Authorities	End Users	Utilize the system for early wildfire detection and informed decision-making in forest management and conservation.
2	Firefighting Services	Operational Stakeholders	Receive early alerts to enable rapid response, containment, and mitigation of wildfire incidents.
3	Local Communities	Indirect Beneficiaries	Benefit from early detection through reduced risk to life, property, and natural resources.
4	Private Landowners	Direct Beneficiaries	Monitor privately owned forests or agricultural land to detect fire incidents early and prevent property and economic loss.
5	Environmental Protection Agencies	Regulatory Stakeholders	Use detection insights to support environmental protection, policy planning, and compliance monitoring.

2.3 User Characteristics

The FIRO wildfire detection system is designed for users with varying technical backgrounds. The expected user categories and their characteristics are described below:

2.3.1 Forest Department Officers

These users are responsible for monitoring forest regions and responding to wildfire incidents. They possess basic computer literacy but may not have advanced technical knowledge of machine learning systems. The system is designed to be intuitive and requires minimal training for effective use.

2.3.2 Firefighting Personnel

Firefighters and emergency response teams use the system to receive early fire alerts and monitor wildfire events. They require fast, reliable, and clear information rather than technical model details. They receive a notification on Whats app or mobile device with accurate location and alert that they can follow immediately.

2.3.3 System Administrators

System administrators are responsible for maintaining the Raspberry Pi hardware, managing the web application, and updating the deployed model. They possess intermediate to advanced technical skills related to system configuration, networking, and software maintenance.

2.3.4 General Users

General users may access the system through the web interface to view detection results or upload images. They are assumed to have basic web navigation skills and no prior knowledge of deep learning or edge computing.

2.4 Requirements Elicitation

Requirements elicitation for the FIRO wildfire detection system was conducted through a structured analysis of the problem domain, stakeholder needs, and technical constraints. The process aimed to identify functional and non-functional requirements necessary to develop a practical, reliable, and deployable wildfire detection solution.

The elicitation process involved reviewing existing wildfire detection literature, analyzing limitations of current monitoring approaches, and examining real-world constraints associated with forest environments in Pakistan and Azad Jammu and Kashmir. Feedback from academic supervisors and potential end users, such as forestry personnel and emergency responders, was also considered to ensure system usability and relevance.

Key requirements were derived by evaluating the operational conditions of forest regions, including limited power availability, restricted internet connectivity, and the need for low-latency detection. Based on these factors, the system requirements emphasize edge-based processing, minimal hardware dependency, and simple user interaction.

Additionally, technical feasibility studies were carried out to assess the computational capabilities of low-power devices such as Raspberry Pi. This analysis influenced the selection of a lightweight deep learning model and the adoption of model optimization techniques to ensure real-time performance.

The final set of requirements reflects a balance between detection accuracy, system efficiency, cost constraints, and ease of deployment. These requirements form the foundation for the system design, implementation, and evaluation phases of the FIRO project.

2.5 Product Functions

The FIRO system is designed as an integrated wildfire monitoring solution that combines data acquisition, intelligent analysis, and user interaction within a single framework. It captures visual information from the environment, prepares it for analysis, and performs automated fire detection directly at the point of deployment. By processing data locally, the system supports timely decision-making and reliable operation even in areas with limited connectivity.

Detection results are stored and synchronized through a cloud-based backend, enabling persistent record keeping and centralized access to wildfire events. A web-based interface allows stakeholders to visualize detection outcomes, review historical records, and monitor system activity. This ensures that relevant authorities and users can easily access and interpret information needed for situational awareness and response planning.

The system follows a modular and extensible design philosophy, allowing it to adapt to future requirements without major redesign. Its architecture supports scalability, integration of additional components, and functional expansion, making it suitable as a foundation for broader wildfire monitoring and early warning applications in real-world settings.

2.5.1 Functional Requirements

Table 2.3 presents the functional requirements of the FIRO wildfire detection system. These requirements describe the expected system behavior and user interactions, defining how the application captures data, processes wildfire detections, presents results, and manages access and storage.

Image Input Handling

The system shall allow continuous image capture through a camera connected to the edge device for real-time environmental monitoring. Additionally, users shall be able to upload images through the web interface to report potential wildfire incidents manually. This functionality supports both automated surveillance and user-assisted fire reporting.

Image Preparation

The system shall automatically prepare all incoming images to ensure they are suitable for wildfire analysis. Image preparation ensures consistency across different input sources and enables reliable processing regardless of image size or format.

Wildfire Detection

The system shall analyze input images to determine the presence or absence of wildfire activity. Detection results shall be generated in a timely manner to support rapid decision-making and early warning.

Edge-Based Processing

The system shall perform wildfire analysis directly on the edge device without requiring continuous cloud-based computation. This enables low-latency operation and ensures functionality in areas with limited or unreliable internet connectivity.

Efficient Model Execution

The system shall utilize optimized inference mechanisms to ensure reliable performance on resource-constrained hardware. This ensures that real-time wildfire detection can be sustained without excessive computational overhead.

Result Visualization

The system shall present wildfire detection outcomes through a web-based interface. Users shall be able to view detection results and confidence indicators in a clear and accessible format to support situational awareness.

Event Recording

The system shall record wildfire detection events along with relevant contextual information such as timestamps and detection outcomes. Event records support system evaluation, historical analysis, and incident review.

User Access Control

The system shall require users to authenticate before accessing monitoring and management features. Authentication ensures that only authorized users can view sensitive information and interact with system controls.

System Status Monitoring

The system shall provide feedback on its operational status, allowing administrators to verify system availability and identify potential faults during operation.

Data Management

The system shall store uploaded images and detection records securely to enable future review, reporting, and analysis while preserving data integrity.

2.5.2 Non-Functional Requirements

This subsection provides a detailed explanation of the non-functional requirements listed in Table **??**. These requirements define the quality attributes, operational constraints, and performance characteristics of the FIRO wildfire detection system, ensuring that it operates efficiently, reliably, and securely in real-world deployment environments.

Table 2.3: Functional Requirements of the FIRO System

Req. ID	**Requirement**	**Description**
FR-01	Image Input Handling	The system shall capture images automatically or accept user-uploaded images through the web interface.
FR-02	Image Preparation	The system shall prepare input images to ensure compatibility with wildfire analysis.
FR-03	Wildfire Detection	The system shall analyze images to identify the presence or absence of wildfire activity.
FR-04	Edge-Based Processing	The system shall perform wildfire analysis locally on the edge device without continuous cloud dependency.
FR-05	Efficient Model Execution	The system shall execute optimized inference to support real-time detection on resource-limited hardware.
FR-06	Result Visualization	The system shall display wildfire detection results through a web-based monitoring interface.
FR-07	Event Recording	The system shall record detection events and associated metadata for monitoring and analysis.
FR-08	User Access Control	The system shall require user authentication to access monitoring and management features.
FR-09	System Status Monitoring	The system shall provide feedback on system availability and operational status.
FR-10	Data Management	The system shall securely store images and detection records for future review and reporting.

Performance

The FIRO system is required to perform real-time wildfire detection with low inference latency on resource-constrained hardware. Model inference must be fast enough to support continuous monitoring without noticeable delay, ensuring timely detection of wildfire events. Achieving real-time performance is essential for early warning and rapid response, particularly in high-risk forest regions.

Accuracy

The system is designed to achieve a classification accuracy of approximately 96% on the wildfire dataset. High accuracy is critical to minimize false positives, which may lead to unnecessary alerts, and false negatives, which may result in delayed response to actual fire incidents. This requirement ensures that the system produces reliable and actionable predictions under normal operating conditions.

Reliability

The FIRO system must operate continuously and maintain stable performance during extended monitoring periods. Reliability ensures that the system remains functional under normal environmental and hardware conditions without frequent failures, crashes, or degradation in detection quality. Continuous operation is essential for long-term forest surveillance and early wildfire detection.

Scalability

The system architecture is designed to support future scalability without requiring major structural modifications. This includes the ability to integrate additional cameras, deploy multiple edge devices, or incorporate alert mechanisms such as SMS or mobile notifications. Scalability ensures that the FIRO system can evolve to meet increasing monitoring demands.

Maintainability

The system is designed with a modular structure to simplify maintenance, updates, and troubleshooting. Maintainability ensures that individual system components, such as the deep learning model or user interface, can be updated or repaired without affecting the overall system operation. This reduces long-term maintenance costs and improves system longevity.

Security

Security is a critical requirement of the FIRO system to protect sensitive detection data and system access. User authentication mechanisms are implemented to restrict access to authorized users only. Stored data and system interactions are protected against unauthorized access, ensuring data integrity and privacy.

Resource Efficiency

The system is optimized to use computational resources efficiently by employing a lightweight and quantized deep learning model. Efficient use of CPU, memory, and power resources ensures stable operation on low-power hardware and enables continuous monitoring without excessive energy consumption.

Availability

The FIRO system is required to maintain high availability, remaining operational during extended monitoring periods with minimal downtime. This ensures that wildfire detection services remain accessible at all times, particularly during peak fire seasons when continuous surveillance is critical.

Chapter 3

System Design

3.1 Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) diagram of the Edge-Based Wildfire Detection System presents a hierarchical decomposition of the project into well-defined and manageable components. At the top level, the project is divided into six major phases: Research, Hardware Setup, Image Processing, Model Training Deployment, Deployment Pilot Testing, and Alert Monitoring System. The research phase focuses on literature review, hardware selection, and model identification. Hardware setup covers Raspberry Pi configuration, camera interfacing, sensor integration, and power/network validation. Image processing includes image capture, resizing and normalization, and feature enhancement for robust inference. The model training and deployment phase involves training the MobileNetV2 model, converting it to TensorFlow Lite, and deploying it on the Raspberry Pi for edge inference. Deployment and pilot testing emphasize controlled site deployment, calibration, and environment testing to validate system reliability. Finally, the alert and monitoring system manages alert configuration, WhatsApp notifications, result uploads to Firebase, and dashboard visualization, ensuring real-time wildfire monitoring and response readiness.

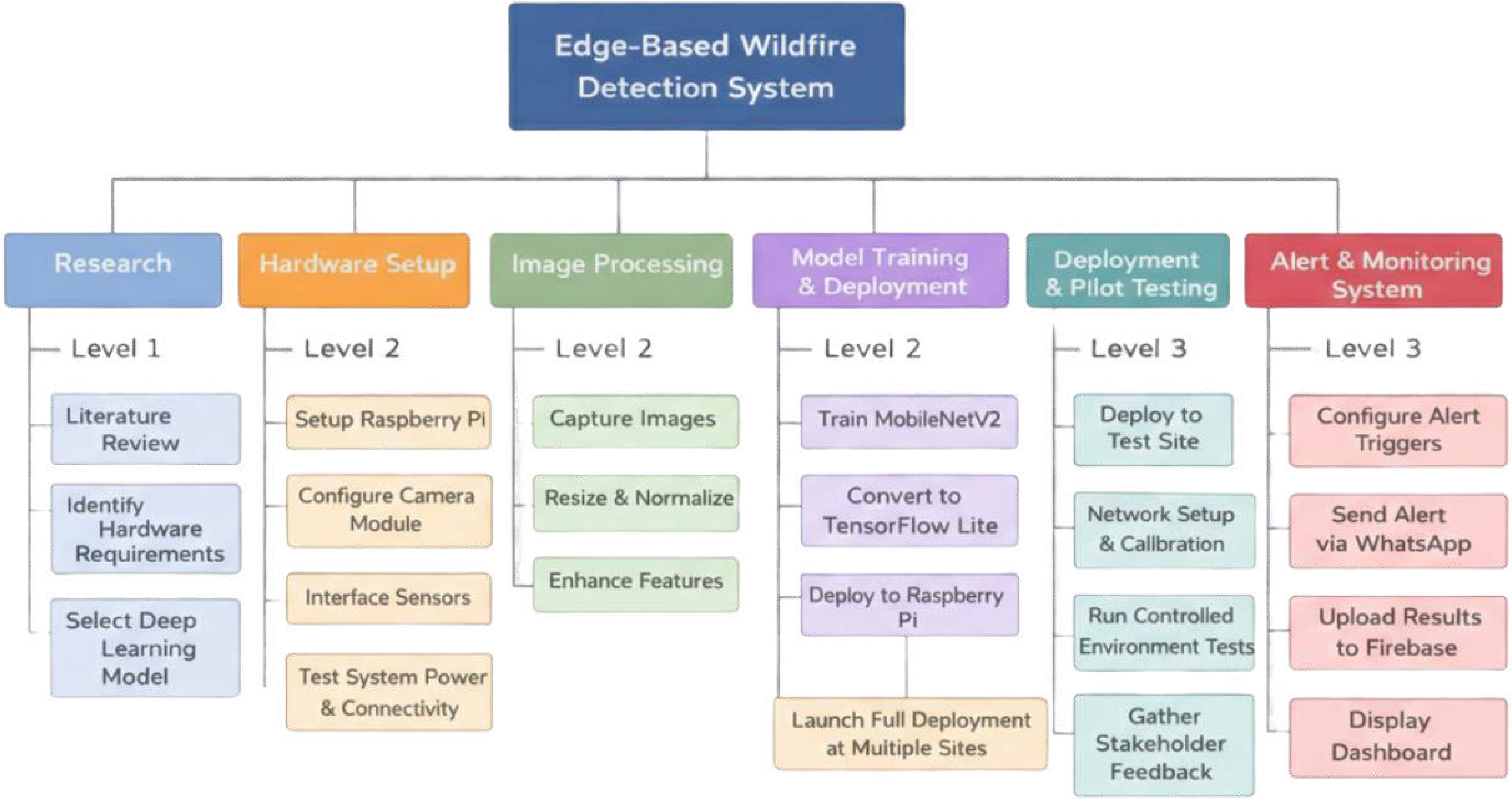

Figure 3.1: Work Breakdown Structure

3.2 Dataset Analysis

3.2.1 Dataset Description

The dataset used in this project consists of RGB images collected from publicly available wildfire image repositories and curated datasets commonly used in forest fire detection research. The primary source of the dataset is an open-access forest fire image collection available on Kaggle, which contains real-world images of forest scenes with and without fire. The dataset includes diverse environmental conditions such as daylight variations, different vegetation types, smoke presence, and background clutter, making it suitable for training and evaluating a robust wildfire detection model.

All images are labeled into two classes: Fire and No Fire. Prior to training, the images were resized to 224×224 pixels to match the input requirements of the MobileNetV2 model. The dataset was divided into training, validation, and testing subsets to ensure fair evaluation and to prevent overfitting. Basic preprocessing and data augmentation techniques were applied to improve generalization. 3.2 illustrates sample images from dataset used for training models. [39]

Figure 3.2: Sample of dataset showing fire and no fire instances.

3.2.2 Data Preprocessing

The dataset used in this project consists of RGB images collected from multiple publicly available online sources and curated according to the project requirements. The images were manually organized into two hard-labeled classes, namely Fire and No Fire, where each class is stored in a separate directory and treated as an independent category during training. All images were resized to a uniform resolution of 224×224 pixels to ensure consistency and compatibility with the MobileNetV2 model architecture.

Table 3.1: Dataset Size and Distribution

Dataset Split	**Fire Images**	**No Fire Images**	**Total Images**
Training Set	1940	2410	4350
Validation Set	328	329	656
Test Set	553	688	1241
Total	**2821**	**3427**	**6247**

3.3 Model Description

3.3.1 MobileNet V2

In this project, a lightweight deep learning model was selected to ensure accurate wildfire detection while maintaining real-time performance on a low-power edge device. The primary model employed for wildfire detection in this project is MobileNetV2, implemented in its lightweight and quantized form to support deployment on resource-constrained edge devices. MobileNetV2 is a convolutional neural network architecture specifically designed for efficient inference on mobile and embedded platforms. Its design prioritizes a balance between accuracy and computational efficiency, making it particularly suitable for real-time applications where memory, processing power, and energy consumption are limited, such as on the Raspberry Pi 5 used in this system.

A key architectural feature of MobileNetV2 is the use of depthwise separable convolutions, which decompose standard convolution operations into depthwise and pointwise convolutions. This significantly reduces the number of trainable parameters and floating-point operations compared to traditional convolutional neural networks, while maintaining strong representational capability. As a result, the model achieves faster inference and lower memory usage, enabling continuous image processing on low-power hardware without requiring external GPU resources.

To adapt the model to the wildfire detection task, MobileNetV2 was fine-tuned on a custom dataset consisting of two hard-labeled classes: Fire and No Fire. Transfer learning was applied by initializing the network with pre-trained ImageNet weights and retraining selected layers on the wildfire dataset to capture domain-specific visual features. Furthermore, the trained model was converted to a TensorFlow Lite INT8 quantized format, which reduced the model size and improved inference speed while incurring negligible loss in classification accuracy. This optimization was essential for achieving real-time performance on the Raspberry Pi 5 and for minimizing energy consumption during continuous operation.

MobileNetV2 was selected over heavier architectures due to its low computational complexity, fast inference capability, and high classification accuracy for binary image recognition tasks. Its proven effectiveness in real-time computer vision applications, combined with seamless deployment support through TensorFlow Lite, makes it an ideal choice for edge-based wildfire detection systems where practical deployability is as important as detection performance.

3.3.2 Model Architecture

The wildfire detection model is built upon the MobileNetV2 convolutional neural network architecture, which is specifically designed for efficient inference on resource-constrained devices. The network accepts input images of size $224 \times 224 \times 3$ $224{\times}224{\times}3$ and begins with an initial standard convolution layer using a 3×3 $3{\times}3$ kernel and stride of 2 to reduce spatial resolution while increasing channel depth. This is followed by a sequence of inverted residual blocks, which form the core of the MobileNetV2 architecture.

Each inverted residual block consists of three main components: a 1×1 1×1 pointwise convolution for channel expansion, a 3×3 3×3 depthwise separable convolution for spatial feature extraction, and a final 1×1 1×1 pointwise linear projection that compresses features back to a lower-dimensional space. Linear bottlenecks are used at the block outputs to preserve information by avoiding non-linear activation at low-dimensional representations. Residual connections are applied between blocks when the input and output dimensions match, improving gradient flow and training stability. The network progressively increases feature depth while reducing spatial resolution, producing a compact yet expressive feature representation. 3.3 illustrates the Mobilenet V2 architecture. [40]

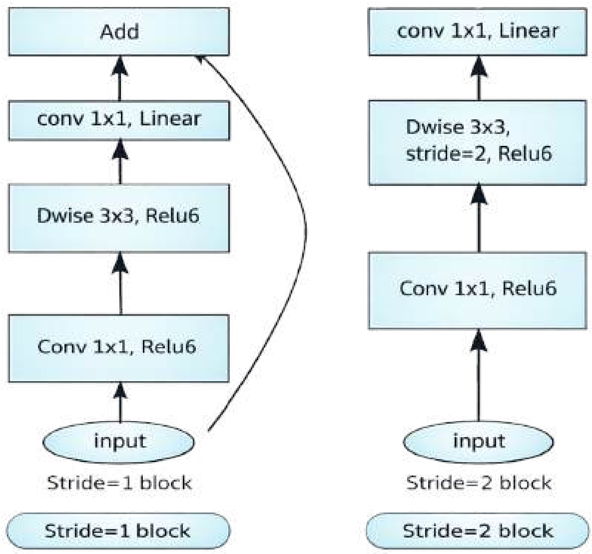

Figure 3.3: Inverted residual blocks used in the MobileNetV2 architecture. The stride-1 block includes a residual connection for feature reuse, while the stride-2 block performs spatial downsampling without skip connections.

After the final inverted residual block, a 1×1 1×1 convolution is applied to expand the feature channels, followed by a Global Average Pooling layer that aggregates spatial information into a single feature vector. This pooled representation is passed to a fully connected dense layer to learn task-specific feature combinations. The output layer consists of two neurons with a Softmax activation function, producing normalized probabilities corresponding to the *Fire* and *No Fire* classes. ReLU activations are used throughout the network except at the linear bottleneck layers, in accordance with the original MobileNetV2 design.

For training, the model uses categorical cross-entropy as the loss function and is optimized using the Adam optimizer. Transfer learning is applied by initializing the backbone with ImageNet-trained weights and fine-tuning selected layers to adapt the feature representations to wildfire imagery. This architecture achieves an effective balance between representational capacity and computational efficiency, making it suitable for real-time inference on embedded edge platforms.

To enable efficient deployment of the FIRO wildfire detection model on resource-constrained edge hardware, the trained MobileNetV2 model was optimized using INT8 post-training quantization and converted into TensorFlow Lite format. Figure **??** illustrates the complete conversion pipeline adopted in this project, highlighting each transformation stage from the original floating-point model to the final edge-optimized deployment model.

The process begins with a fully trained MobileNetV2 model using 32-bit floating-point (FP32) weights. This model is first serialized into a standard TensorFlow SavedModel or `.h5` format, which serves as the input for the TensorFlow Lite conversion process. To ensure accurate quantization, a representative calibration dataset consisting of real wildfire and non-fire images is provided. This dataset is used to statistically estimate activation ranges, enabling precise scaling of both weights and activations during quantization.

Subsequently, INT8 post-training quantization is applied, reducing numerical precision from FP32 to 8-bit integers. This step significantly decreases model size, memory footprint, and computational complexity while preserving detection accuracy. The TensorFlow Lite Converter then generates a fully quantized *MobileNet V2 Lite* model, optimized for low-latency inference. The final quantized model is deployed using the TensorFlow Lite Interpreter on Raspberry Pi, enabling real-time, energy-efficient wildfire detection entirely at the edge without reliance on cloud-based computation.

3.3.3 Training Methdology

The wildfire detection model was developed and trained using a transfer learning strategy based on the MobileNetV2 architecture. All experiments were conducted in a GPU-accelerated environment to efficiently handle the computational requirements of deep convolutional neural networks. Training was performed using Python 3.11 with TensorFlow 2.x and Keras as the primary deep learning frameworks, which provide high-level APIs for model construction, optimization, and evaluation. The use of an NVIDIA Tesla P100 GPU significantly reduced training time and enabled rapid experimentation with hyperparameters and fine-tuning strategies. For deployment, the trained model was optimized and executed on a Raspberry Pi 5, representing a realistic low-power edge computing platform suitable for continuous wildfire monitoring.

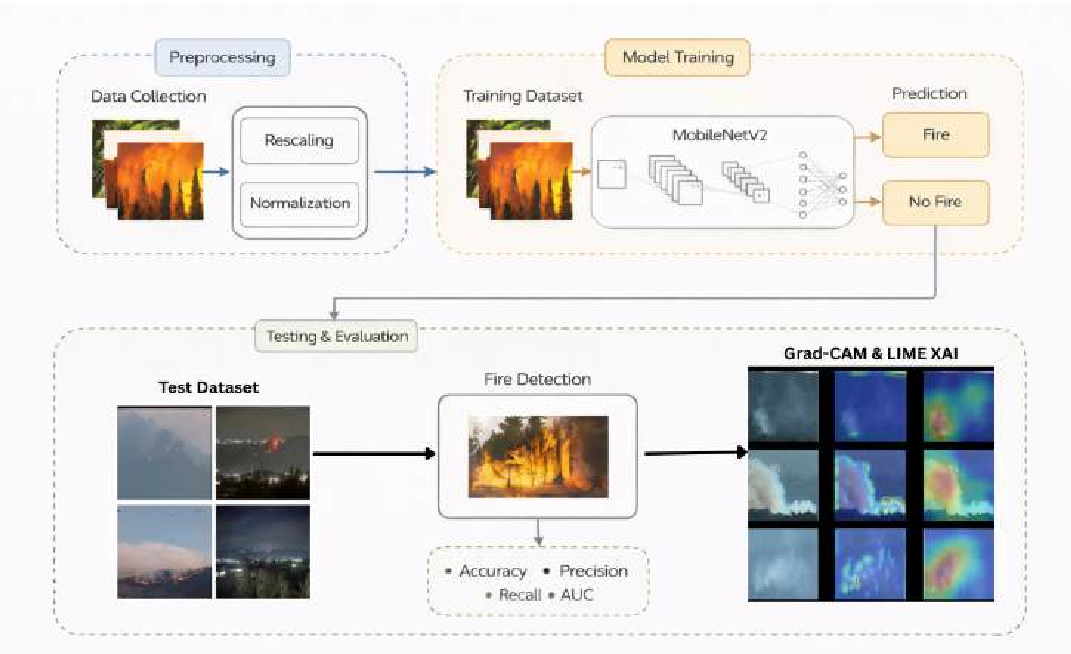

Figure 3.4: End-to-end workflow of the FIRO wildfire detection system. The pipeline illustrates data collection and preprocessing, MobileNetV2-based model training, prediction, system evaluation using standard performance metrics and explainable AI.

3.3.4 Activation Functions and Loss Function

ReLU (Rectified Linear Unit) activation functions are employed throughout the convolutional layers of the MobileNetV2 backbone due to their ability to introduce non-linearity while maintaining computational efficiency and mitigating vanishing gradient problems. In the final classification layer, a Softmax activation function is used to convert raw model outputs into normalized class probabilities, enabling probabilistic interpretation of predictions. The model is trained using the Binary Cross-Entropy loss function, which is well-suited for binary classification problems where the objective is to distinguish between two mutually exclusive classes. This loss function penalizes incorrect predictions proportionally to their confidence, encouraging the model to produce well-calibrated probability estimates.

3.3.5 Hyperparameter Configuration and Training Strategy

The training process utilized an input image size of $224 \times 224 \times 3$ with a batch size of 32, balancing convergence stability and GPU memory utilization. The Adam optimizer was selected due to its adaptive learning rate mechanism and robustness across a wide range of deep learning tasks. During the initial feature extraction phase, the learning rate was set to 0.01 while keeping the pretrained MobileNetV2 base layers frozen, allowing the newly added classification layers to adapt to the wildfire dataset. After achieving stable convergence, a fine-tuning phase was initiated by unfreezing the last layers of the base network and retraining them with a reduced learning rate of 1×10^{-5}. This gradual unfreezing strategy enabled the model to learn domain-specific wildfire features while minimizing the risk of catastrophic forgetting and overfitting.

3.3.6 Evaluation Metrics and Model Optimization

Model performance was evaluated using multiple metrics, including accuracy, precision, and recall. Accuracy provides an overall measure of correct predictions, while precision and recall offer deeper insight into the model's ability to correctly identify wildfire events and minimize false alarms. AUC was included to assess the model's discriminative capability across varying classification thresholds. To ensure efficient deployment on edge hardware, the trained floating-point model was converted into a TensorFlow Lite INT8 quantized format using post-training quantization. This process reduced model size, memory footprint, and inference latency while maintaining competitive accuracy. The resulting quantized model enables real-time wildfire detection on the Raspberry Pi 5, demonstrating the practical viability of the proposed FIRO system for continuous, low-power forest monitoring.

3.4 Raspberry Pi–Based Hardware Development and Configuration

The hardware part of the FIRO wildfire detection system is focused on a Raspberry Pi, which serves as the main edge-computing unit to acquire images in real-time, execute inferences and communicate the data. The Raspberry Pi has been chosen because it is low cost, small form factor, has enough computing power and extensively supported by camera modules and networking interfaces, which is why it is applicable in remote and resource constrained forest applications.

The Raspberry Pi is connected to a camera module directly by the CSI (Camera Serial Interface) port to record visual data in the area the camera is focused on, which is the forest. The camera will constantly take images at set intervals or on command such as to monitor the situation, as per the monitoring setup. The images are directly sent to the local processing pipeline which is operating on the Raspberry Pi and external data transfer is not required at the first steps of detection. This design guarantees low latency and also interruption-free functionality when the network connectivity is problematic or restricted.

Table 3.2: Hyperparameter Configuration for MobileNetV2 Wildfire Detection Model

Hyperparameter	**Value / Description**
Base Architecture	MobileNetV2 (Pretrained on ImageNet)
Input Image Size	$224 \times 224 \times 3$
Batch Size	32
Optimizer	Adam
Initial Learning Rate	0.01 (Feature extraction phase)
Fine-Tuning Learning Rate	1×10^{-5}
Loss Function	Binary Cross-Entropy
Activation Function (Hidden Layers)	ReLU
Activation Function (Output Layer)	Softmax
Number of Output Classes	2 (Fire, No Fire)
Training Strategy	Transfer learning with frozen base layers followed by selective fine-tuning
Evaluation Metrics	Accuracy, Precision, Recall, AUC
Quantization Method	Post-training INT8 quantization
Deployment Format	TensorFlow Lite (MobileNet V2 Lite)
Training Hardware	NVIDIA Tesla P100 GPU
Deployment Hardware	Raspberry Pi 5

After an image is taken, Raspberry Pi is used to perform all the preprocessing tasks. This involves image resizing to a fixed resolution, normalization and formatting it as per the needs of deployed deep learning model. The image is passed through the preprocessing after which it passes to the optimized inference engine executing on the machine. The model can run locally on the TensorFlow Lite runtime, which enables itself to make efficient and fast predictions without having to use cloud-based computing. This edge processing method is very useful in offering shorter response time and improving system reliability in case of the vital wildfire incidences.

After the inference, the Raspberry Pi analyses the result of the prediction and comes up with a detection value, which is either a fire or no-fire condition. With the classification outcome, other metadatas like the date and rough location data is also added to the detection event. This information is subsequently ready to be sent to the cloud layer to be logged and visualized.

In cloud communication, Raspberry Pi makes use of its inbuilt networking functionality, either Ethernet or Wi-Fi connections to send the detection outcomes to Firebase cloud database safely. Just lightweight metadata and inference output is relayed which consumes the least bandwidth but gives real-time alignment with the monitoring dashboard. Such a design provides local autonomy and centralized visibility that enables the authorities to obtain detection information remotely without overloading the edge device.

In general, the hardware setup allows FIRO to be an intelligent edge node that includes all the components to monitor wildfires continuously. A combination of local image processing, on-device inference and selective cloud synchronization will allow optimization of performance, scalability, and practical deployability of the system in a real world forest monitoring system.

3.5 Use Case Description

This section describes the primary use cases of the FIRO wildfire detection system. The use cases define how different users interact with the system to perform wildfire monitoring, detection, and management tasks. The system is designed to be simple and intuitive, allowing both technical and non-technical users to operate it effectively.

Use Case 1: User Login

Actor: Authorized User (Forest Officer, Firefighter, Administrator)
Description: This use case allows authorized users to access the system by providing valid login credentials. Authentication ensures that only permitted users can view detection results and manage system operations.
Precondition: User must be registered in the system.
Postcondition: User is successfully authenticated and redirected to the dashboard.
Main Flow:

- User enters username and password.
- System validates credentials.
- User gains access to the main dashboard.

Use Case 2: Image Capture / Upload

Actor: Detection Unit (Raspberry Pi 5 with Camera)

Description: This use case describes the automatic image acquisition process in which the system periodically captures RGB images using a camera mounted with the Raspberry Pi 5 for wildfire detection. No manual image upload or web-based input is involved.

Precondition: The camera is properly connected to the Raspberry Pi 5 and the system is powered on with scheduled capture enabled.

Postcondition: The captured image is temporarily stored in the system memory and forwarded for preprocessing and model inference.

Main Flow:

- The system automatically triggers the camera at predefined time intervals.
- The camera captures an RGB image of the monitored forest area.
- The system validates the captured image resolution and format.
- The image is forwarded to the preprocessing module for resizing and normalization.

Use Case 3: Wildfire Detection

Actor: System
Description: The system processes the input image and performs fire or no-fire classification using the MobileNetV2 deep learning model deployed on the Raspberry Pi.
Precondition: Valid RGB image is available.
Postcondition: Detection result is generated with confidence score.
Main Flow:

- Image is resized and normalized.
- MobileNetV2 model performs inference.
- Classification result is produced.

Use Case 4: View Detection Results

Actor: Authorized User
Description: This use case allows users to view wildfire detection results through the web interface, including prediction label and confidence score.
Precondition: Detection process is completed.
Postcondition: Results are displayed on the dashboard.
Main Flow:

- System displays fire/no-fire result.

- Confidence score is shown to the user.

Use Case 5: Event Logging

Actor: System

Description: The system records wildfire detection events, including timestamps and prediction outcomes, for monitoring and future analysis.

Precondition: Detection result is generated.

Postcondition: Event data is stored in the system database.

Main Flow:

- Detection event is logged.
- Data is saved for historical reference.

Use Case 6: System Administration

Actor: System Administrator

Description: This use case allows the administrator to manage system configuration, update the deployed model, and maintain system availability.

Precondition: Administrator authentication.

Postcondition: System configuration or model is updated successfully.

Main Flow:

- Administrator accesses system settings.
- Performs maintenance or model update.
- System resumes normal operation.

Use Case 7: Public Fire Incident Reporting

Actor: Common User

Description: This use case allows members of the public to report suspected wildfire incidents by uploading an image and providing the geographic location through the web interface.

Precondition: Public access to the reporting web page and availability of an internet connection.

Postcondition: The reported fire image and location details are stored in the cloud database for review and further action.

3.6 Use Case Diagrams

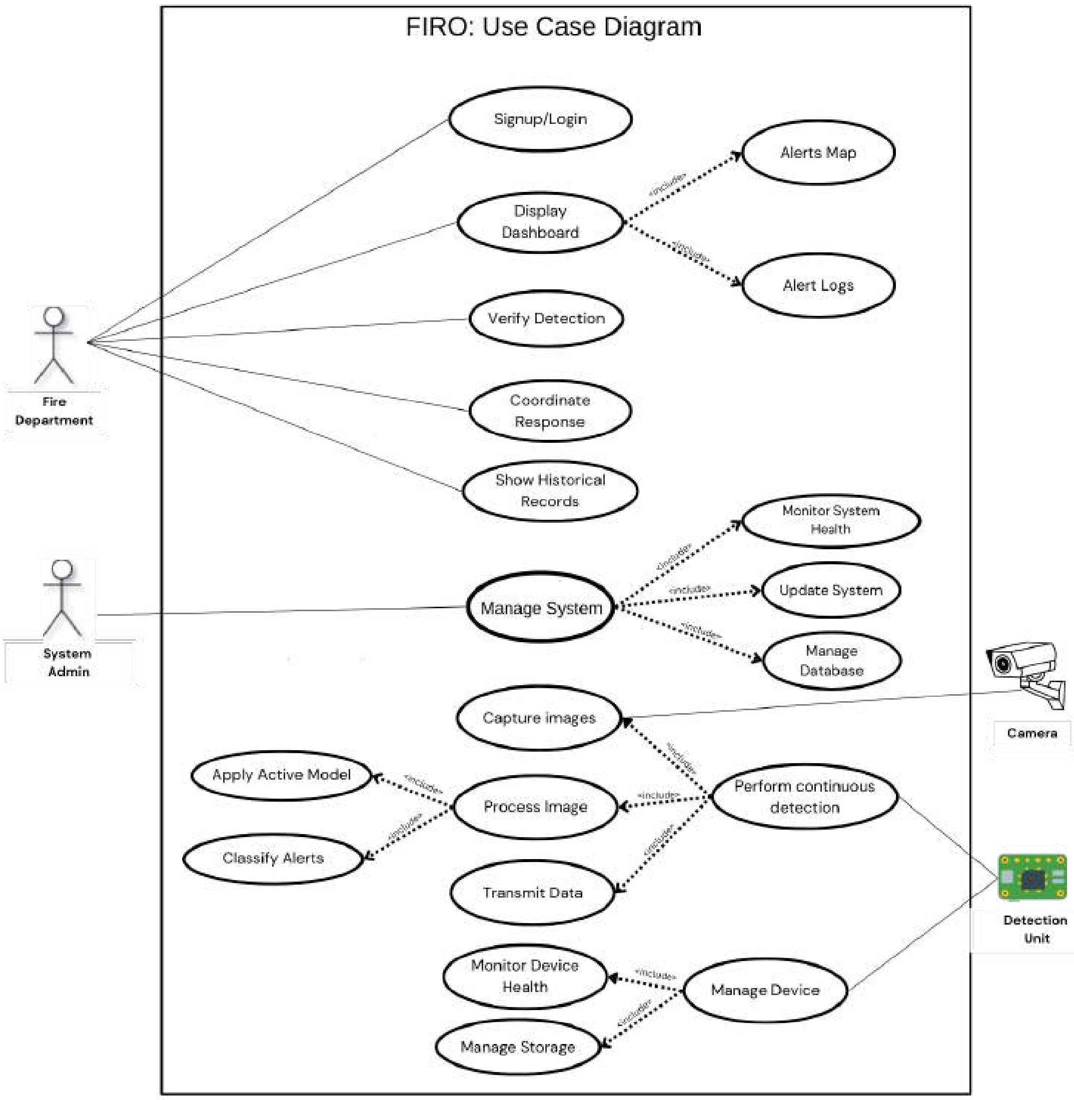

Figure 3.5: Use Case Diagram

3.7 Activity Diagram

This activity diagram illustrates the workflow followed by the Fire Department after receiving a wildfire alert in the FIRO system. The process starts when an alert is delivered via WhatsApp, prompting officials to access the fire alerts dashboard for further details. Using the dashboard, the fire location is identified on an interactive map, after which the presence of wildfire is verified. A decision point determines whether further verification is required; if confirmed, the response is coordinated immediately, otherwise the verification loop continues. The activity concludes once the appropriate response actions have been initiated or the alert is validated as non-critical.

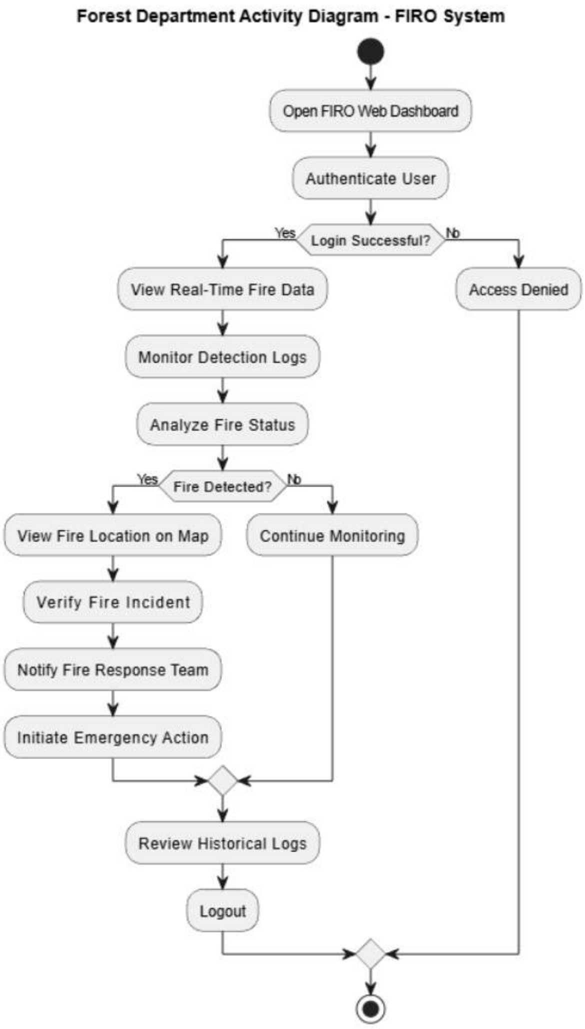

Figure 3.6: Activity Diagram - Fire Department Interaction

3.8 Sequence Diagram

3.8.1 Forest Department Sequence Diagrams

This sequence diagram illustrates the login interaction of the Forest Department with the FIRO system. The process begins when the fire department user enters their username and password, which are forwarded by the edge server to the cloud database for authentication. Upon successful validation, the user profile is retrieved from the database, and the edge server grants access to the system by loading the dashboard, enabling the fire department to monitor alerts and wildfire-related information.

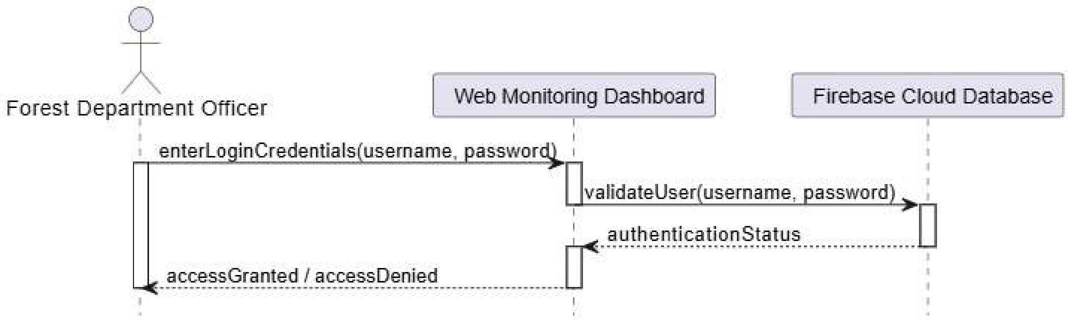

Figure 3.7: Sequence Diagram - Login

Forest Department Data Retrieval

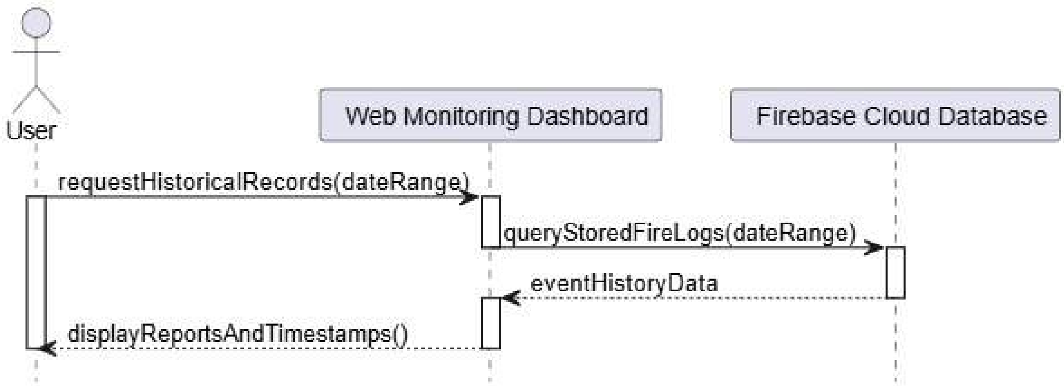

Figure 3.8: Data Retrieval Sequence Diagram

Historical Records

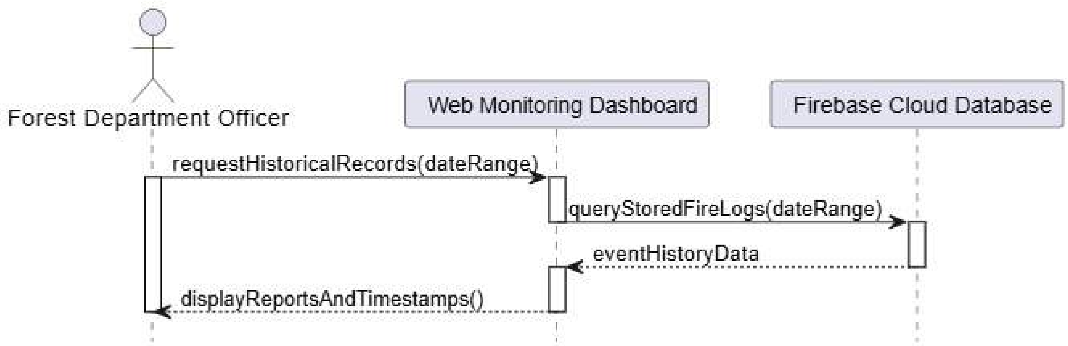

Figure 3.9: Historical Records Retrieval Sequence Diagram

3.8.2 Common User

The Common user will access web site and report a incident he can attach a photo and live location of fire incident that data will be stored in cloud and evaluated through model for a quick action the user can also access historical data.

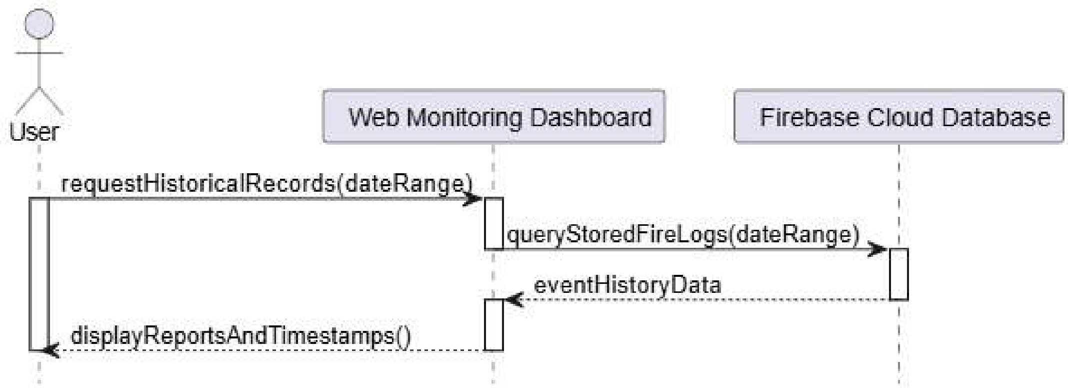

Figure 3.10: Common User Sequence Diagram

3.9 System Architecture

The FIRO system architecture illustrates an end-to-end edge-based wildfire detection pipeline designed for efficiency and real-time response. A tower-mounted

RGB camera continuously captures forest images, which are processed locally on a Raspberry Pi 5 where images are resized to 224×224 RGB format and analyzed using a lightweight, INT8-quantized MobileNetV2 (TensorFlow Lite) model. Instead of transmitting raw images, only compact prediction metadata—including fire/no-fire classification, confidence score, geographic location, device ID, and timestamp—is sent over the internet to the Firebase cloud database, significantly reducing bandwidth usage and latency. Firebase acts as a central coordination layer, storing alerts and triggering real-time notifications via WhatsApp to Forest Department staff, while simultaneously updating a web-based dashboard managed by a Forest Department manager for monitoring incidents, reviewing logs, and coordinating responses.

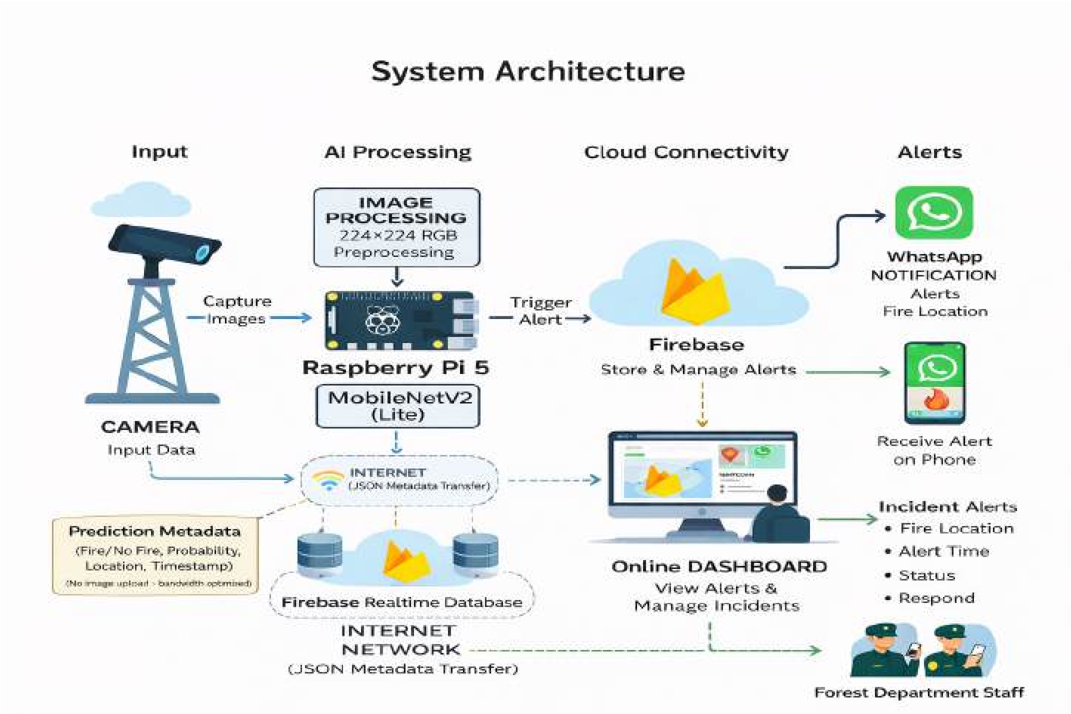

Figure 3.11: FIRO System Architecture

Chapter 4

System Testing

4.1 Test Cases

This table is a summary of functional test cases that will be utilized in validating the workflow of the FIRO system of image acquisition up to the generation of alerts. The functional test cases are used to test a key component of the system such as image capture, preprocessing, model inference, cloud data storage, alert triggering, and dashboard visualization. All the tests were conducted successfully and it confirmed that the system is working in the proper way and is reliable. 4.1 shows the test cases and expected results.

4.2 Unit Testing

A unit testing was done in a systematic manner to determine the correctness, reliability and robustness of each of the modules in the FIRO wildfire detection system before complete integration of the system. This testing stage was aimed at isolating the components and testing their functionality separately to reduce the possibility of cascading errors in the course of end-to-end functionality.

Table 4.1: Test Cases

Test ID	**Description**	**Expected Result**
1	Automatic image capture using Raspberry Pi camera	Image is captured successfully
2	Image preprocessing (resize to 224×224 RGB)	Image is resized and normalized
3	Model inference using MobileNetV2 Lite	System predicts Fire or No Fire
4	Upload prediction result to Firebase database	Data is stored in Firebase
5	Trigger WhatsApp alert on fire detection	Alert message is sent
6	Display prediction on web dashboard	Fire/No Fire status is shown

The camera module was also tested to ensure that it acquired images continuously and automatically in a fixed time duration to ensure that there is a constant communication between the Raspberry Pi and the camera hardware. The image preprocessing module was tested to verify proper resizing of images that were captured. 224×224 224x224 pixels, support of RGB channels, and normalization of pixel values that the MobileNetV2 model demands. The model inference module was strictly tested on test samples of fire and non fire images to test whether the model can successfully discriminate between binary and confidence score all under edge- device constraints.

Moreover, the database communication module was tested with the purpose to ensure that the results of the prediction, timestamps and device identifiers, as well as the location metadata are properly formatted and transmitted to Firebase cloud database. The alert generation component was also tested separately to ensure that WhatsApp alert is only generated after positive fire detection, thus eliminating false and unnecessary alert. The successful completion of all the unit tests prove that each of the modules is functioning according to the desired purpose and provides a steady backbone upon which a stable and reliable operation of the system-level can be ensured.

4.3 Real Environment System Testing

A special evening test experiment was performed in a controlled set- up to test the strength of the FIRO system during the time of low visibility. The main aim of this test was to evaluate the capability of the system to capture false alarms of events related to wildfires at night when there is less illumination and other visual issues like low contrast, artificial lighting and smoke dispersion become more visible.

In this controlled experiment 12 RGB photographs were taken at night with the camera that was connected to the Raspberry Pi of the camera at different angles and distances. The images that were captured were of visible fire flames and smoke patterns, which were illuminated by other surrounding light sources to depict a realistic night-time scene. These images have been directly run through the INT8-quantized MobileNetV2 TensorFlow Lite model which is loaded onto the Raspberry Pi 5 inference has been done completely on the edge device without having to call cloud-computation.

Figure 4.1: Actual forest fire incidents captured using a mobile camera in the surrounding forest areas of Khuiratta and Nakyal cities, Azad Jammu and Kashmir, during December 2025 and January 2026.

The experimental outcomes indicated that the FIRO system was able to classify with 100 percent accuracy and all the fire and smoke in the test images were being detected correctly by the system. This successful operation under an articulated night-time environment proves the stability and durability of the proposed system to monitor the wildfires on a 24-hour round-the-clock basis. The outcomes also confirm the system to be applicable in the actual implementation settings where rapid detection is mandatory under any type of lighting. 4.2 illustrates the model performance on incident pictures in different locations, different times and conditions

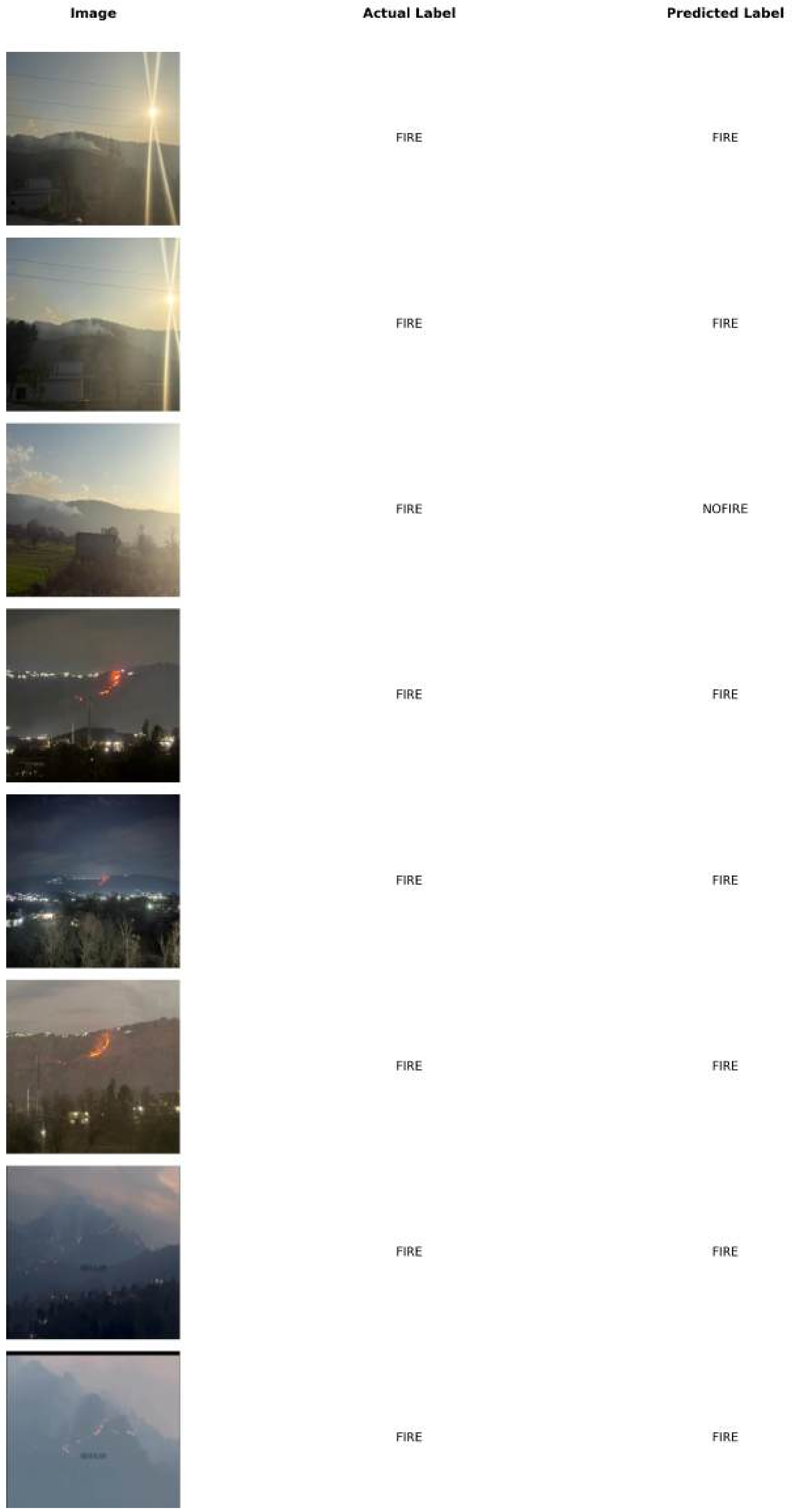

Figure 4.2: Model Testing using Actual forest fire incidents captured using a mobile camera in the surrounding forest areas of Khuiratta and Nakyal cities.

Chapter 5

Results

5.1 Models Comparision

Several architecture models of deep learning were trained and tested to determine the most appropriate architecture to use in detection of wildfires within accuracy and deployment limits. They were VGG-16, ResNet-50, EfficientNet-B0, EfficientNetV2-B0, YOLOv11 Nano (classification variant) and MobileNetV2. The training of all models was employed through a supervised learning method on the same labeled wild fire dataset of two classes Fire and No Fire. All CNN-based architectures used transfer learning by using ImageNet-pretrained weights, which greatly enhanced the rate of convergence and generalization. All the models were tuned with the help of selected hyperparameters, and their performance was measured with the help of the validation and test datasets in order to compare it fairly.

According to the results of the experiment, EfficientNet-B0 has the greatest test accuracy of 98.47 percent, with EfficientNetV2-B0 (97.58 percent) and MobileNetV2 (97.50 percent) coming next. ResNet-50 also had high performance with a validation accuracy of around 96.6 percent and YOLOv11 Nano performed competitively with an accuracy of 96.4 percent and much fewer parameters. Conversely, VGG-16 though with deep architecture and a large number of parameters, achieved a relatively lower performance (93.23 percent) and higher cost of computation. These findings serve to note that the state-of-the-art lightweight architectures are capable of being better than deeper legacy ones as long as they are tuned to specific tasks.

Despite the fact that there were a few models that slightly better than MobileNetV2 in regard to raw accuracy, MobileNetV2 is still chosen as the finally deployed model in the FIRO system because it has the best balance of accuracy, model size, and computational efficiency. MobileNetV2 was able to achieve nearly state-of-the-art accuracy with its 2.6 million parameters, and has a high level of resource-friendliness, making it viable on edge-based inference on resource-constrained hardware, including the Raspberry Pi 5. Moreover, the compatibility of the architecture with TensorFlow Lite and INT8 quantization allowed making large cuts in the amount of memory used and inference latency, which made the architecture the most feasible and risk-averse option in case of real-time wildfire detection in an edge computing setup.

Table 5.1: Performance Comparison of Evaluated Wildfire Detection Models

Model	**Total Parameters**	**Test Accuracy (%)**
MobileNetV2 (Proposed)	2.6 M	97.50
VGG-16	15.24 M	93.23
ResNet-50	∼25 M	96.59
EfficientNet-B0	4.38 M	98.47
EfficientNetV2-B0	6.25 M	97.58
YOLOv11 Nano (Classification)	1.53 M	96.40

5.2 Evaluation Metrics

The quality of the suggested FIRO edge-based wildfire detector was measured quantitatively, based on conventional classification measures usually used in binary image classification exercises. The choice of these metrics was to give a holistic and credible evaluation of the capability of the model to differentiate between the scenarios of Fire and No Fire, especially when it is subjected to real time inference over resource constrained edge hardware. Since wildfire detection is a safety-critical task, the assessment focuses on the detection accuracy in general, as well as the capability of the model to reduce false alarms and missed fires.

One of the main performance measures was accuracy which was applied to determine the overall accuracy of the system. It is the proportion of irregularly classified samples (fire and no-fire) to the total number of reviewed pictures. Although accuracy can give an overall picture of model performance, it cannot be used to give enough information in situations where there can be imbalanced class distributions or where the cost of misclassifications is not equal. However, the high accuracy proves the efficiency of the MobileNetV2-based model in acquiring discriminatory features that are useful in wildfire detection.

Precision was also used to determine the effectiveness of fire predictions the system gives. It calculated the percentage of true fire detections of all samples with predictions of a fire. The FIRO system is especially delicate in terms of precision because fake fire alert can result in unjustified notifications, disruptions in operations and squandering of resources by the forest departments and emergency response teams. With a high precision value, the system makes the events of a detected fire very reliable and useful.

The concept of recall, also known as sensitivity, is an indicator of how effectively the system can detect real instances of wildfires. It is determined as the percentage of correct fire cases which have been correctly identified by the model. Recall is a very important measure in the situation of wildfire monitoring since a missed detection of an actual fire may lead to delayed reaction and devastating environmental and economic outcomes. This is demonstrated by a high recall value that means that the FIRO system is efficient in reducing the occurrence of false negatives and capturing most wildfire occurrences.

F1-score was taken as a balanced score that integrates both the precision and recall into one. Being the harmonic mean of the precision and recall, the F1-score offers a sound assessment of the system when the false positives and the false negatives have to be correctly regulated. This measure is especially appropriate to the system of firo where attentive detection and prompt response are as well crucial. An F1-score value of high is a guarantee that the proposed model has a good balance between reliability of detection and sensitivity, which make it suitable in real-life wildfire detection on edge devices.

5.3 Model Performance

Table 5.2: Performance Comparison of MobileNetV2 Models

Model	Accuracy	Precision	Recall	F1-score
MobileNetV2	0.96	0.96	0.96	0.96
MobileNetV2 Lite	0.95	0.95	0.95	0.95

5.4 Train Vs Validation Accuracy and Loss Graph

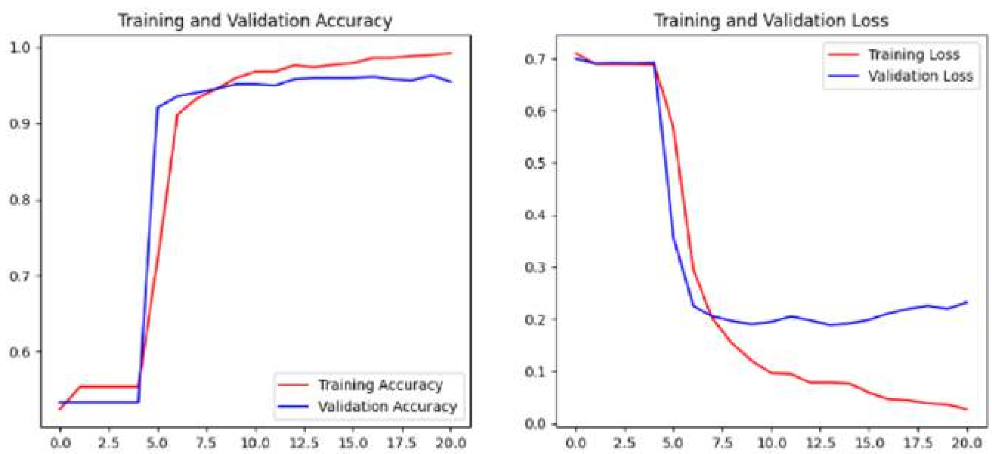

Figure 5.1: Training vs Validation Accuracy and loss

5.5 confusion Matrix

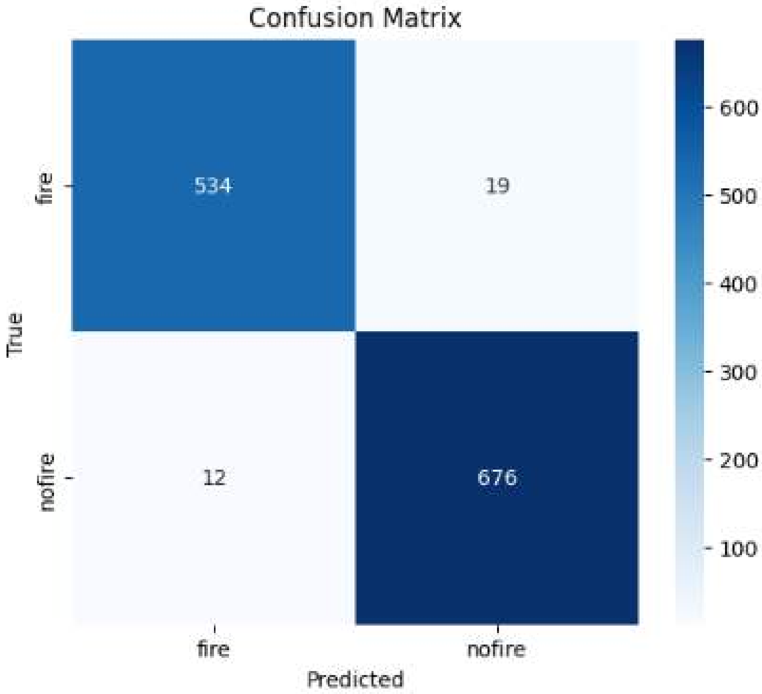

Figure 5.2: Confusion Matrix of the Best Model

Chapter 6

Application Frontend

6.1 Dashboard User Interface

6.1.1 Login Screen

The Login Screen is also the secure access to the FIRO dashboard by the authorized users. It also asks users to use their registered email address and password to open any system features. This authentication system will be used to make sure that sensitive wildfire detection data and monitoring tools are restricted to authorized officers, including forest and fire department officials. Other regular usability features like the ability to recover passwords and the ability to save sessions are also on the screen to make the user more comfortable without affecting the security of the screen.

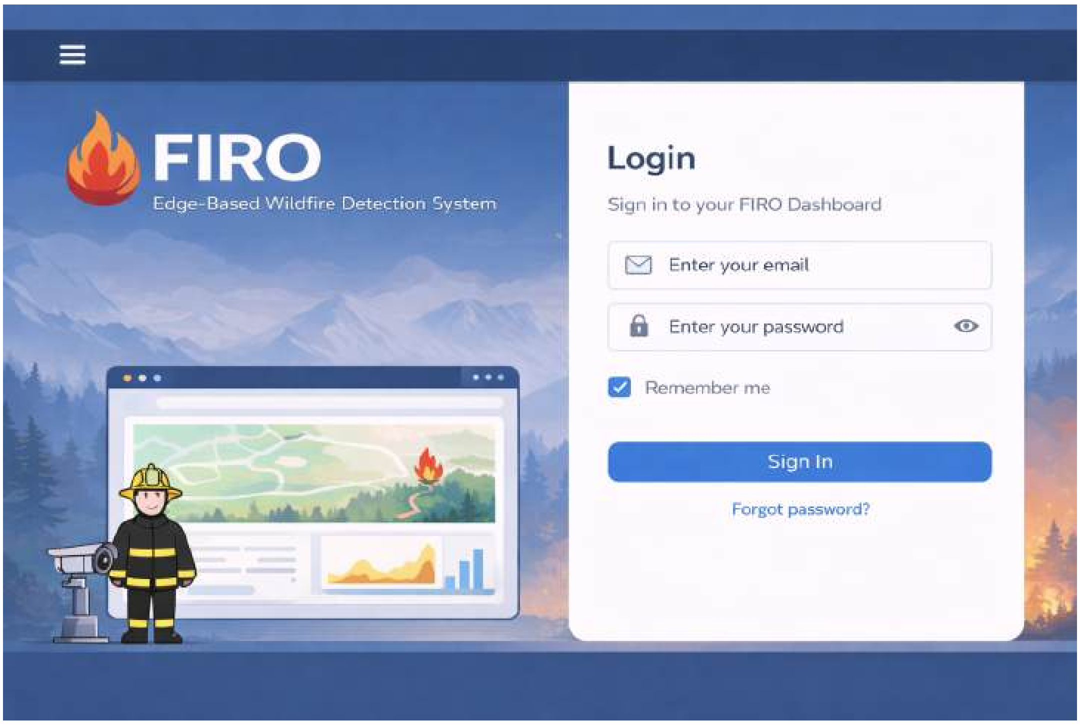

Figure 6.1: Dashboard - Authentication Screen

6.1.2 Dashboard Home Screen

The Dashboard Home Screen gives a general view of the operation of the wildfire monitoring system. It shows major system statistics such as recent detection operations, camera status, and the real-time alerts. The dashboard includes an interactive map that can visually show the locations of detection to enable the user to have a quick look at the spatial distribution of wildfire incidents. This centralized interface allows to have an efficient situational awareness and facilitates timely decision-making when performing monitoring operations.

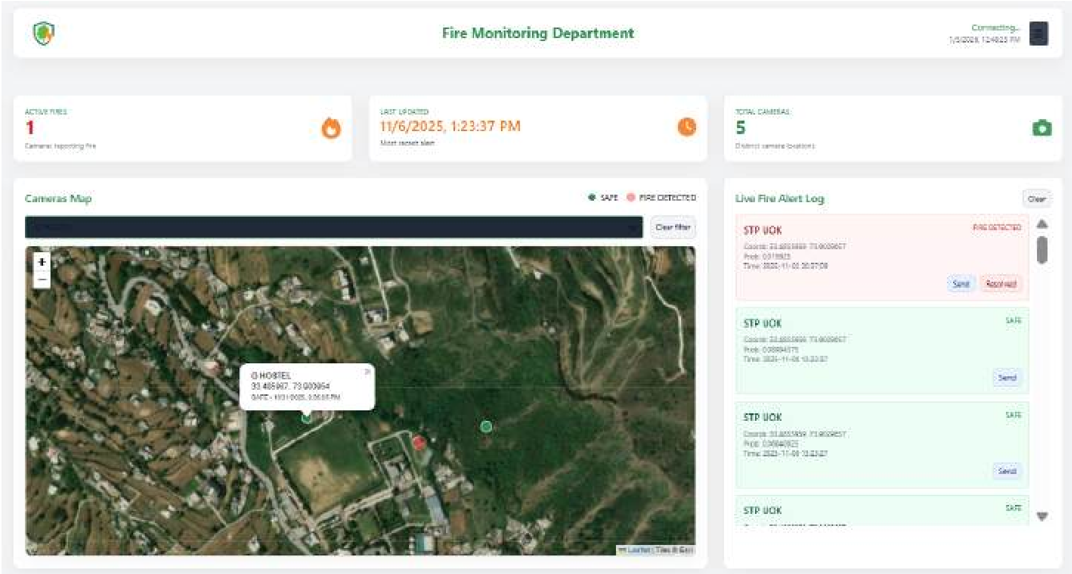

Figure 6.2: Dashboard Main Page

6.1.3 Event Alert Screen

The Event Alert Screen displays real time instances of wildfires identified by the system. The necessary information that is contained in every alert entry is the location coordinates and detection probability, time and the status of the fire that is detected or safe at that point. The user is able to do direct actions which include forwarding alerts to responding authorities or indicating that an event has been solved. This is a prioritized screen that is to be used in active fire situations where there is a need to coordinate rapid response to critical incidents.

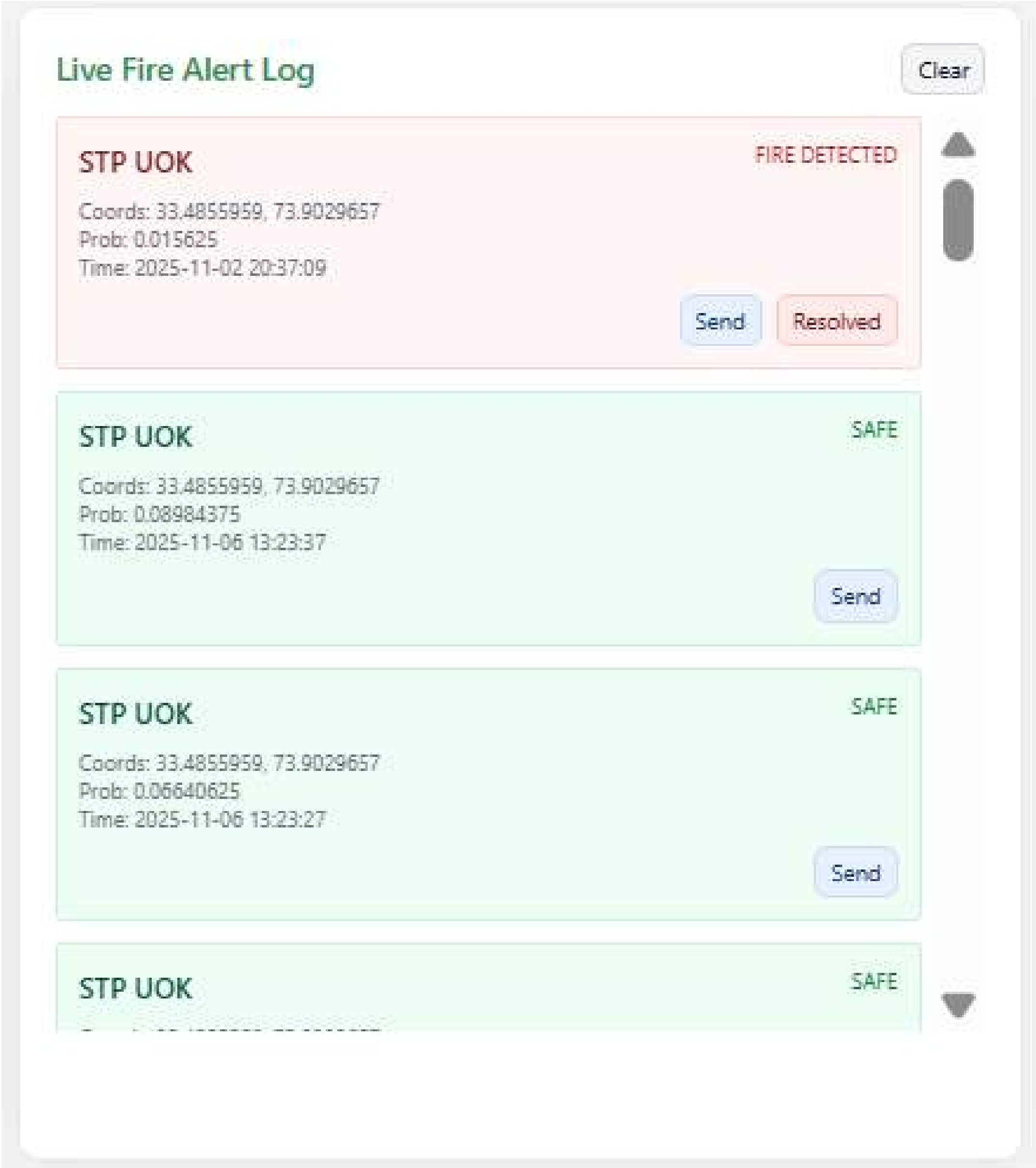

Figure 6.3: FIRO - Event Alert Screen

6.1.4 Personal Settings Screen

Dark Mode interface helps to minimize the visual load and enhance reading when visual conditions are low, e.g. when monitoring at night. This mode allows longer use without any fatigue because of the darker color schemes used without reducing contrast and clarity. Field operators and staff in the control room who would be tracking activity of the wildfires during the night hours will find dark mode useful.

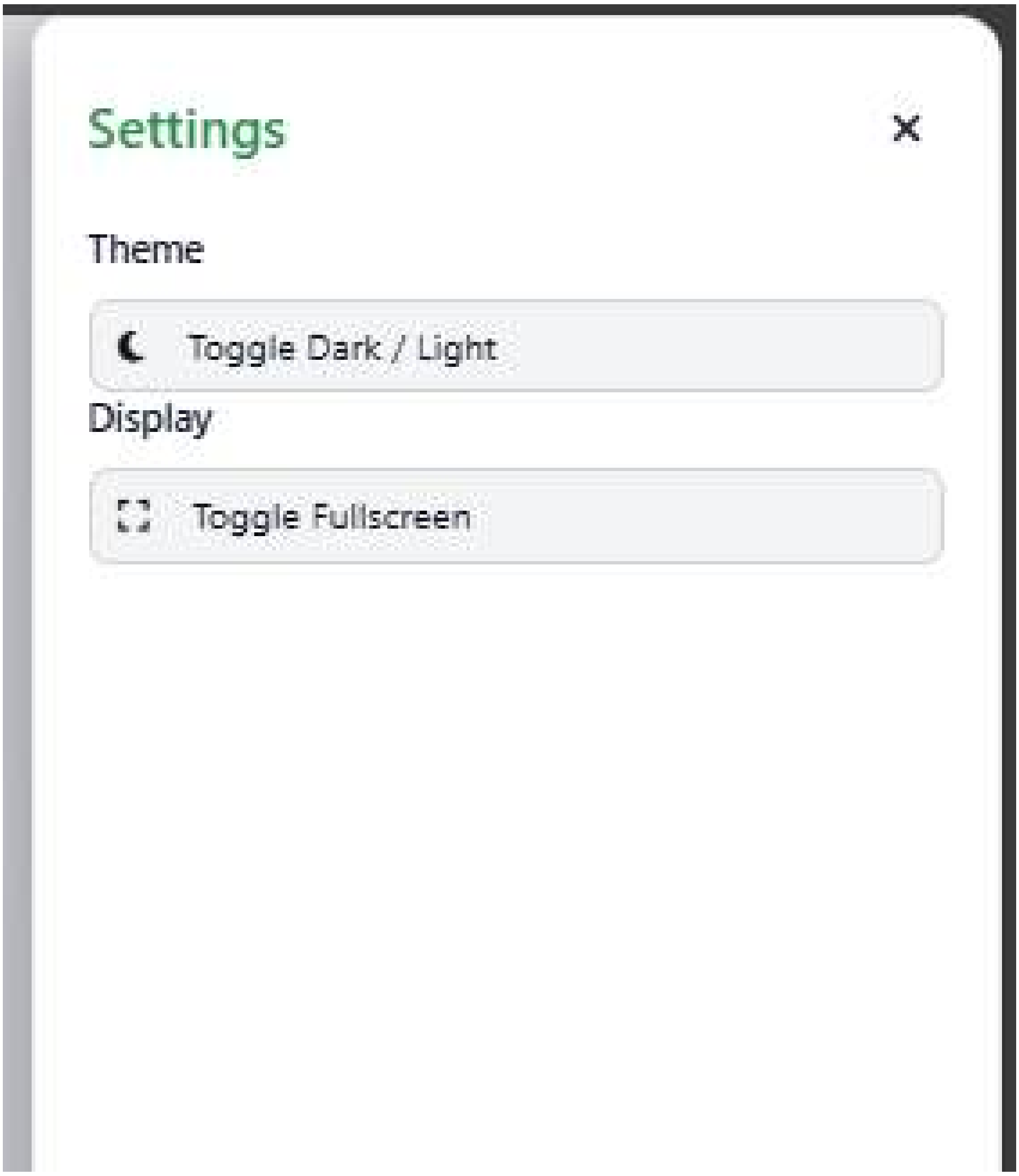

Figure 6.4: FIRO- Dashboard Personal Settings

6.1.5 Event Log History Page

The Event Log History Page is the page that gives a centralized display of all the wildfire detecting events that have been captured by the system. It shows details of events like timestamps of events, whether it was a fire or a safe event, and the identification of the camera or location. The logs can be filtered by the event type to enable users to isolate fire and non-fire events to analyze them. This page allows the authorities to check the past events and confirm the system decision and monitor the wildfires according to time. In general, it facilitates accountability, auditing, and informed decision-making by means of organised event records.

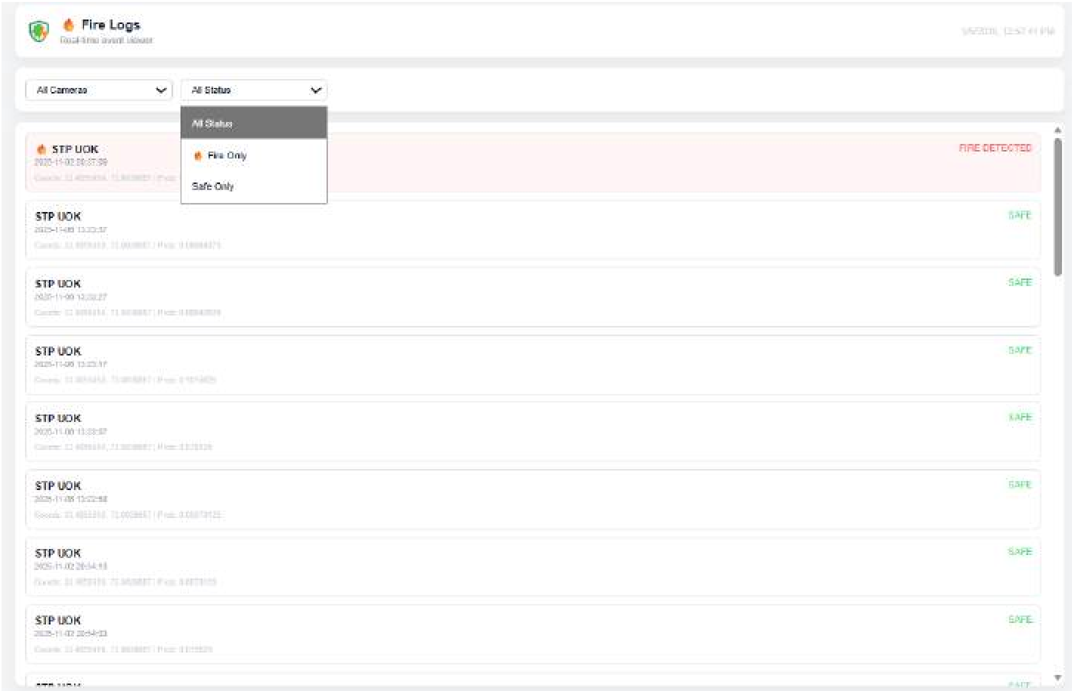

Figure 6.5: Event Log History Page

Chapter 7

Conclusion

7.1 Problems Faced and Lessons Learned

The technical and practical difficulties were encountered during the process of development of the system of detecting wildfires. The solutions to these tasks presented effective insights and learning opportunities, which led to the successful implementation of the project.

7.1.1 Limited Computational Resources on Edge Device

The important issue was the difficulty in implementing a deep learning model on a low-power edge device, like the Raspberry Pi 5. The standard deep learning models are computationally costly and cannot be used in real-time with resource-constrained hardware.

Lesson Learned: It was addressed with the help of choosing a lightweight architecture (MobileNetV2) and through various model optimization methods, including fine-tuning and TensorFlow Lite quantization. It was shown that model efficiency and optimization are important in the case of working with edge devices.

7.1.2 Accuracy vs. Efficiency Trade-off

The other problem was the accuracy of high detection with a smaller model size and inference time. A performance speed was also enhanced through model quantization at the expense of a minimal accuracy loss. We found out that the best combination between precision and efficacy is critical to a real world system. Even a small compromise in accuracy can be forgiven provided that it would allow it to be performed in real time and be practically used.

7.1.3 Environmental Variability

Model predictions were sensitive to the lighting conditions, complexity of the background, and the visibility of the smoke and hence false detections occurred occasionally. This emphasized the need to have diversity in datasets and powerful preprocessing methods. To enhance model generalization, data augmentation and selective dataset are very essential.

7.2 Project Summary

In this project, an edge-based wildfire detection system based on deep learning was designed and implemented. The training was of a MobileNetV2 model with a single objective of classifying forest images as fire and no-fire and with an objective of optimizing deployment on a Raspberry Pi 5 with the help of TensorFlow Lite. Images are automatically captured by the system using a camera and processed locally and the prediction results are sent to a Firebase cloud database. Alerts are also issued and shown on a web dashboard in a case of a fire detection and sent to interested authorities. Experimental evidence shows that the proposed system can be very accurate and perform at real-time and with minimal computational overhead, thus, is appropriate in resource-constrained environments.

7.3 Policy Guidelines

The proposed wildfire detection system shall be developed in line with ethical and policy considerations pertaining to the environmental monitoring systems. Personal and sensitive user data is not gathered by the system and does not violate privacy as the system is based on publicly visible forest areas. The images obtained are automatically processed and not under human supervision; using all the images collected solely to detect fire. The system is also aimed at assisting the early warning and disaster management authorities and not to substitute the human decision-making. The alerts that the system issues are informative and need to be confirmed by the relevant authorities before intervention.

7.4 Future Work

Though it is possible to note that the proposed system shows good performance when it comes to detecting wildfires, a number of improvements can be considered in the future. The system may be enhanced by adding other sensors like the temperature, humidity or gas sensors to enhance early detection. Image acquisition using drones can also be integrated to increase the coverage of vast forest cover. Moreover, more sophisticated methods, including the temporal modeling, domain adaptation, and federated learning, could be considered to enhance the strength and scale. The other area that future studies can concentrate on is large-scale deployment and long-term field testing of the system to further confirm the performance of the system in the real-life situation.

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